1. AI Scoring Inputs (Data Sources)

The CredFlow Risk Engine pulls real-time data from the following four pillars to calculate the **CredScore**:

- **Credit Bureau (CIBIL/CRIF):** Historical repayment patterns, credit utilization, and hard inquiries.
- **GSTN Integration:** Analysis of **GSTR-3B** (Monthly Returns) and **GSTR-1** (Sales) to verify business turnover and tax consistency.
- **Bank Statement Analyzer (BSA):** Scanning the last 6 months of PDF/e-statements for Average Monthly Balance (AMB), cheque bounces, and recurring EMI obligations.
- **Digital Footprint:** Transaction velocity on UPI, POS machines, and e-commerce platforms.

2. The CredScore Hierarchy (Risk Tiers)

- The engine assigns a proprietary score between **300 and 900**. This score dictates the final interest rate offered to the customer.

CredFlow Risk Assessment & Pricing Logic

- **Risk Category: Platinum**
  - **CredScore Range:** 780 – 900
  - **Interest Rate Multiplier:** Base Rate (e.g., 9%)
  - **Approval Logic:** Green Channel (Instant Approval)
- **Risk Category: Gold**
  - **CredScore Range:** 700 – 779
  - **Interest Rate Multiplier:** Base + 2%
  - **Approval Logic:** Auto-Approval (Post-KYC)
- **Risk Category: Silver**
  - **CredScore Range:** 650 – 699
  - **Interest Rate Multiplier:** Base + 4%
  - **Approval Logic:** Conditional (Requires 12-month Bank Statement)
- **Risk Category: Sub-Prime**
  - **CredScore Range:** Below 650
  - **Interest Rate Multiplier:** Base + 7%
  - **Approval Logic:** Manual Underwriting Required

3. Automatic Rejection Triggers (Red Flags)

The AI will automatically issue a "Decline" status if any of the following are detected:

1. **Cheque Bounces:** More than 2 inward cheque/NACH bounces in the last 3 months.
2. **Tax Default:** Any active litigation or non-filing of GST returns for more than 2 consecutive quarters.
3. **Recent Defaults:** Any "Written Off" or "Settled" status on a loan within the last 24 months.
4. **Leverage Ratio:** Total monthly EMI obligations exceeding 60% of the applicant's net monthly income.

4. MSME Specific Risk Modifiers

For business loans (**MSME Growth**), the engine applies these specific weights:

- **Vintage:** Businesses operating for >3 years receive a **+50 point** boost.
- **Industry Risk:** Sectors like "Manufacturing" and "Essential Services" receive lower risk weights than "Hospitality" or "Speculative Trading."
- **GST Consistency:** A variation of >20% in declared sales month-over-month triggers a manual "Revenue Stability" audit.

5. KYC & Fraud Detection

- **Face-Match:** Cross-references the live selfie with the PAN/Aadhaar card photo.
- **Geo-Tagging:** Verifies that the applicant’s current location matches the stated business/residential address.
- **Liveness Check:** Prevents spoofing using AI-generated video or static photos.